

SwasthyaSetu AI

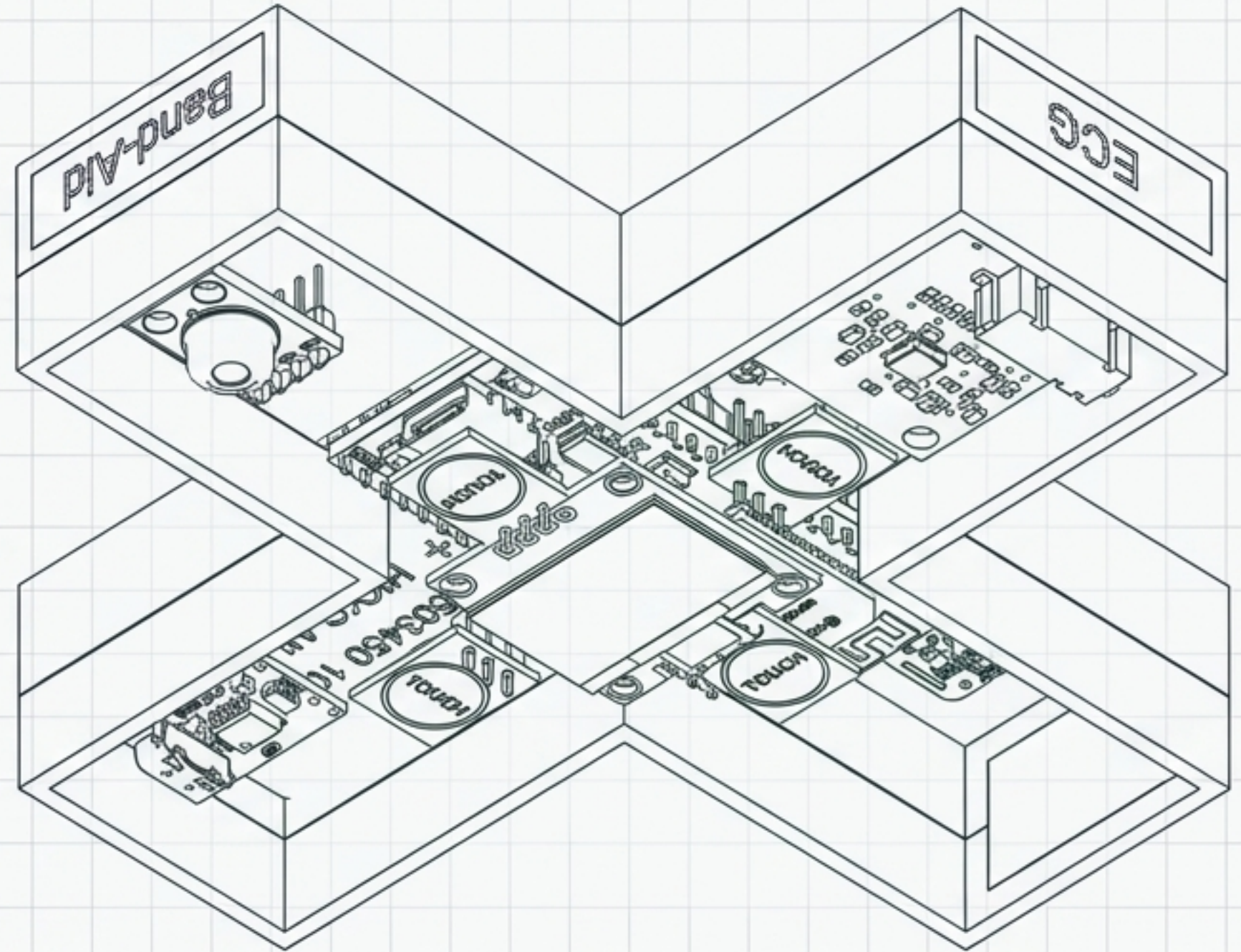
System Architecture, Process Flows, and Engineering Blueprint

Problem Statement ID: 26181

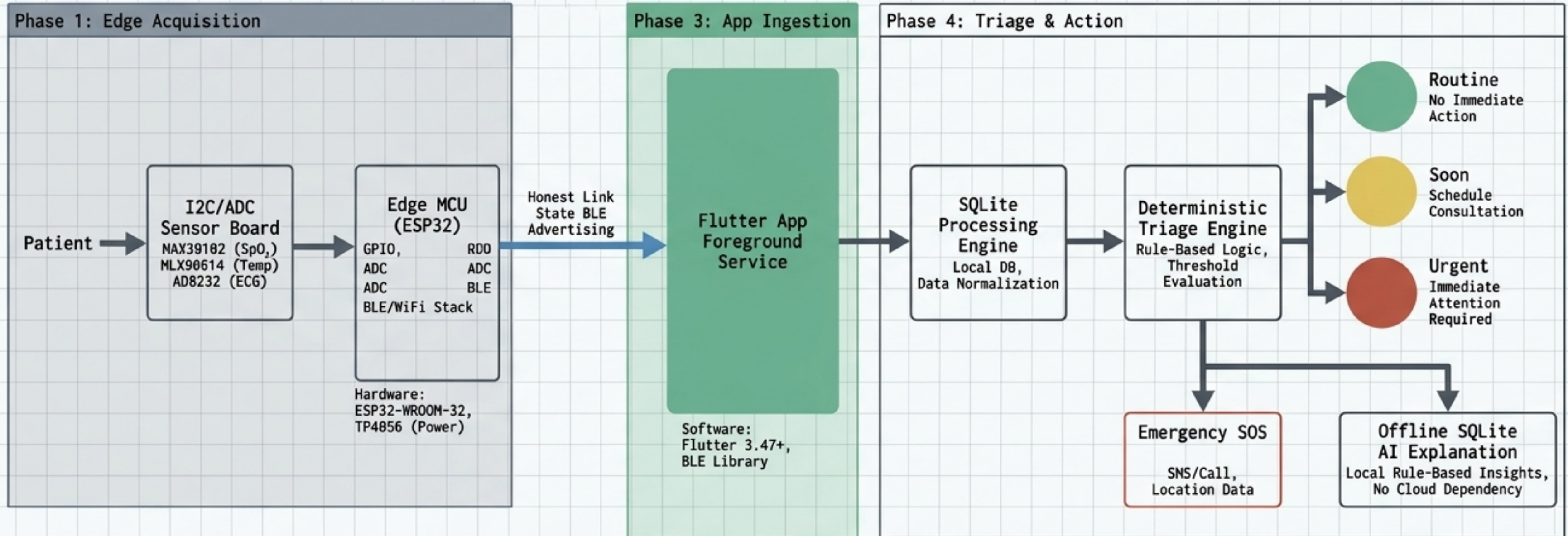
Theme: MedTech / BioTech / HealthTech

Category: Hardware

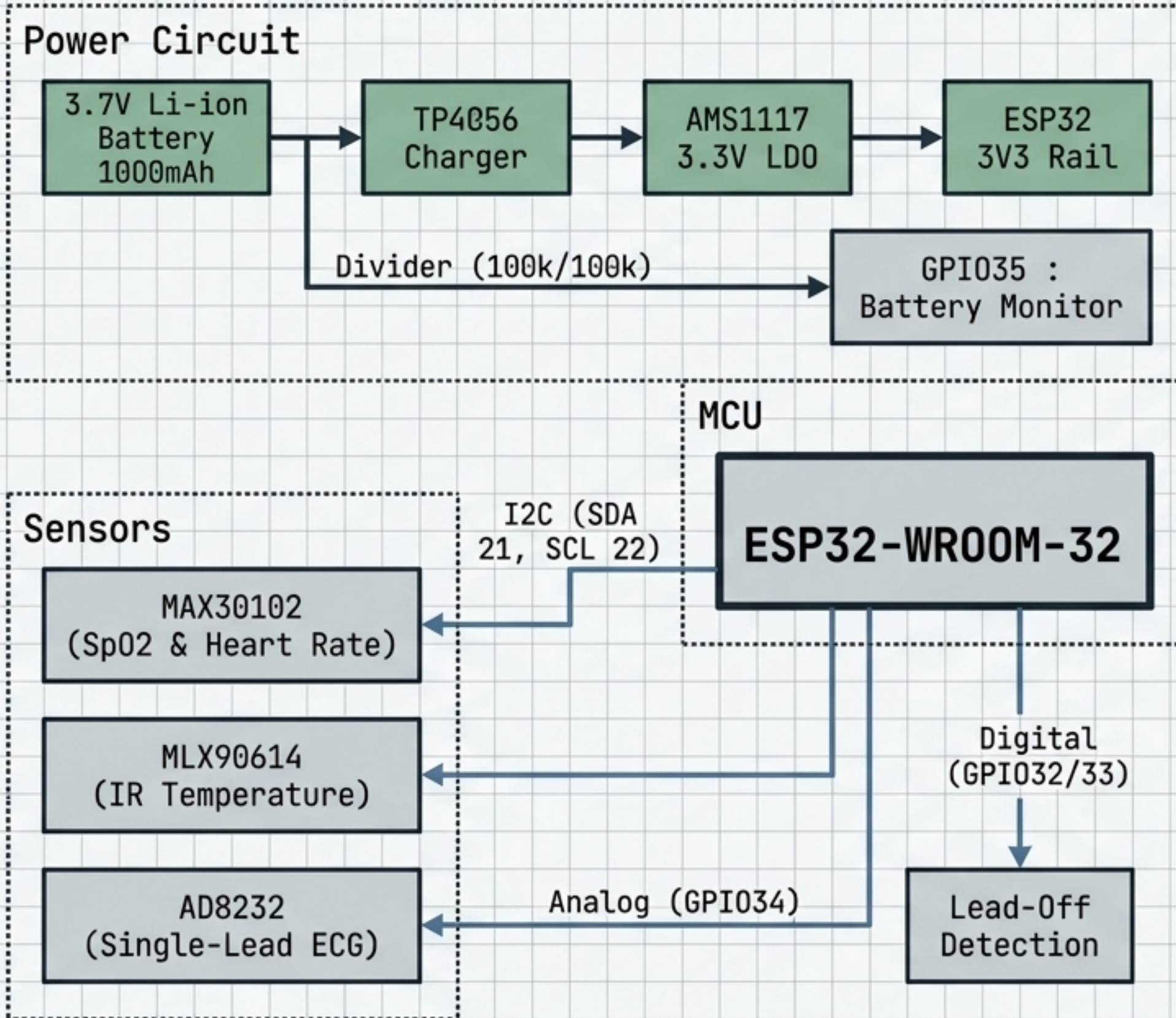
Status: Offline-First Zero-Dependency



Macro System Architecture: End-to-End Signal Lifecycle



Phase 1: Hardware Block Diagram & Pin Mapping



Pin Mapping Reference

I2C Bus (Shared):

- SDA on GPIO21 (4.7kΩ pull-up)
- SCL on GPIO22 (4.7kΩ pull-up)
- Address 0x57 (MAX30102)
- Address 0x5A (MLX90614)

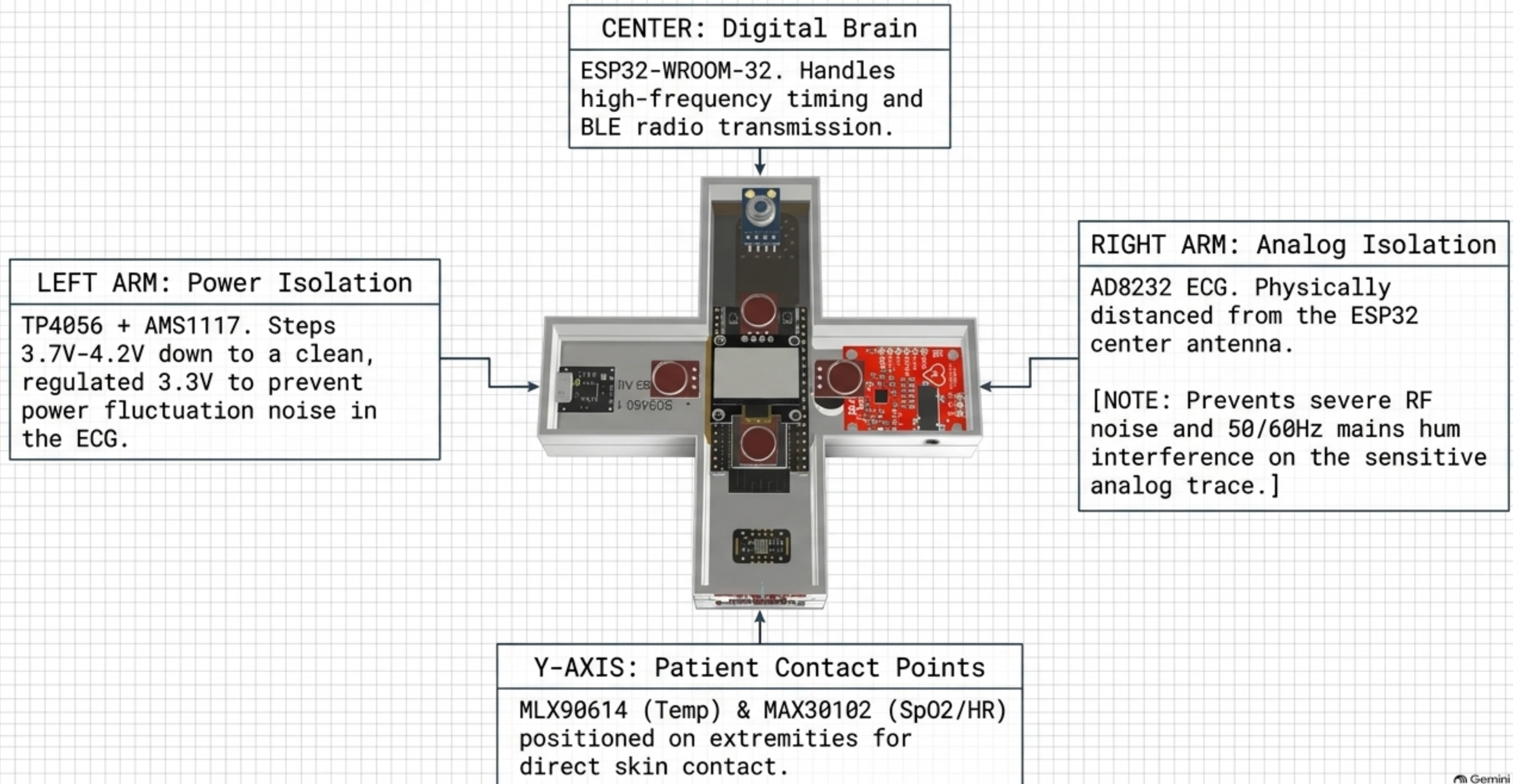
Analog:

- AD8232 Output on GPIO34 (ADC1_CH6)

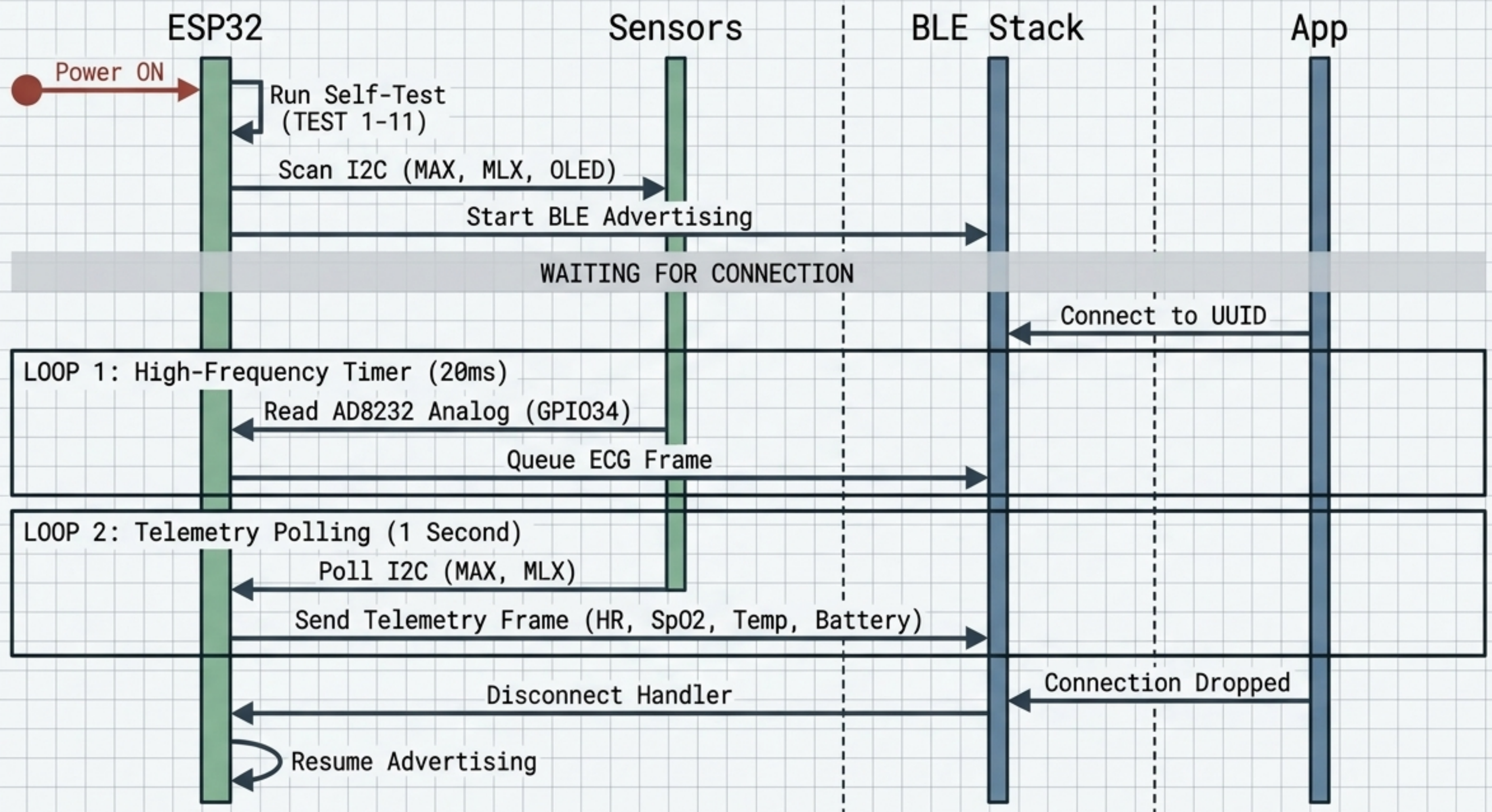
Battery ADC:

- GPIO35 (ADC1_CH7)

Physical Board Layout & Signal Isolation Strategy



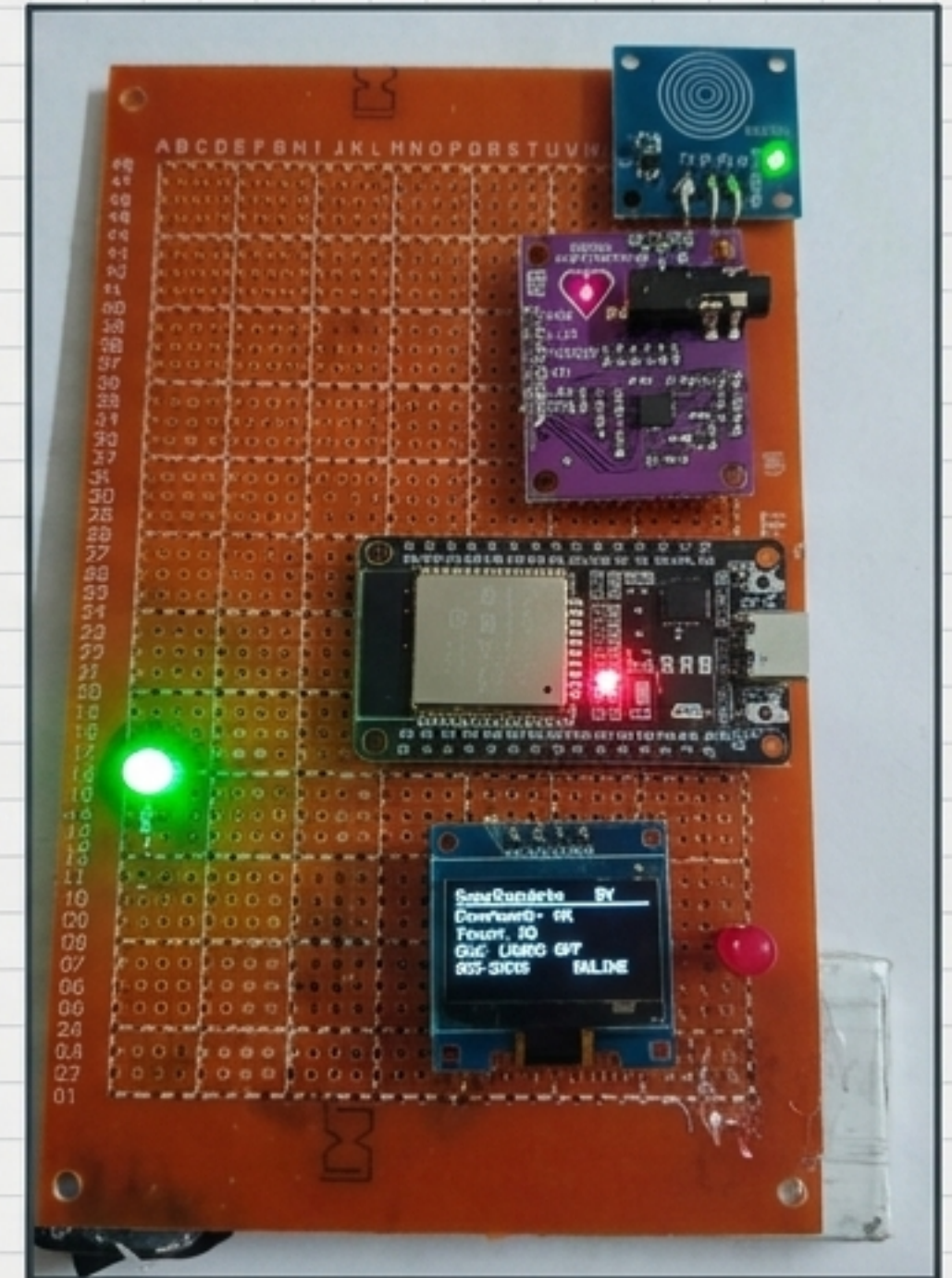
Firmware Execution & Data Polling Sequence



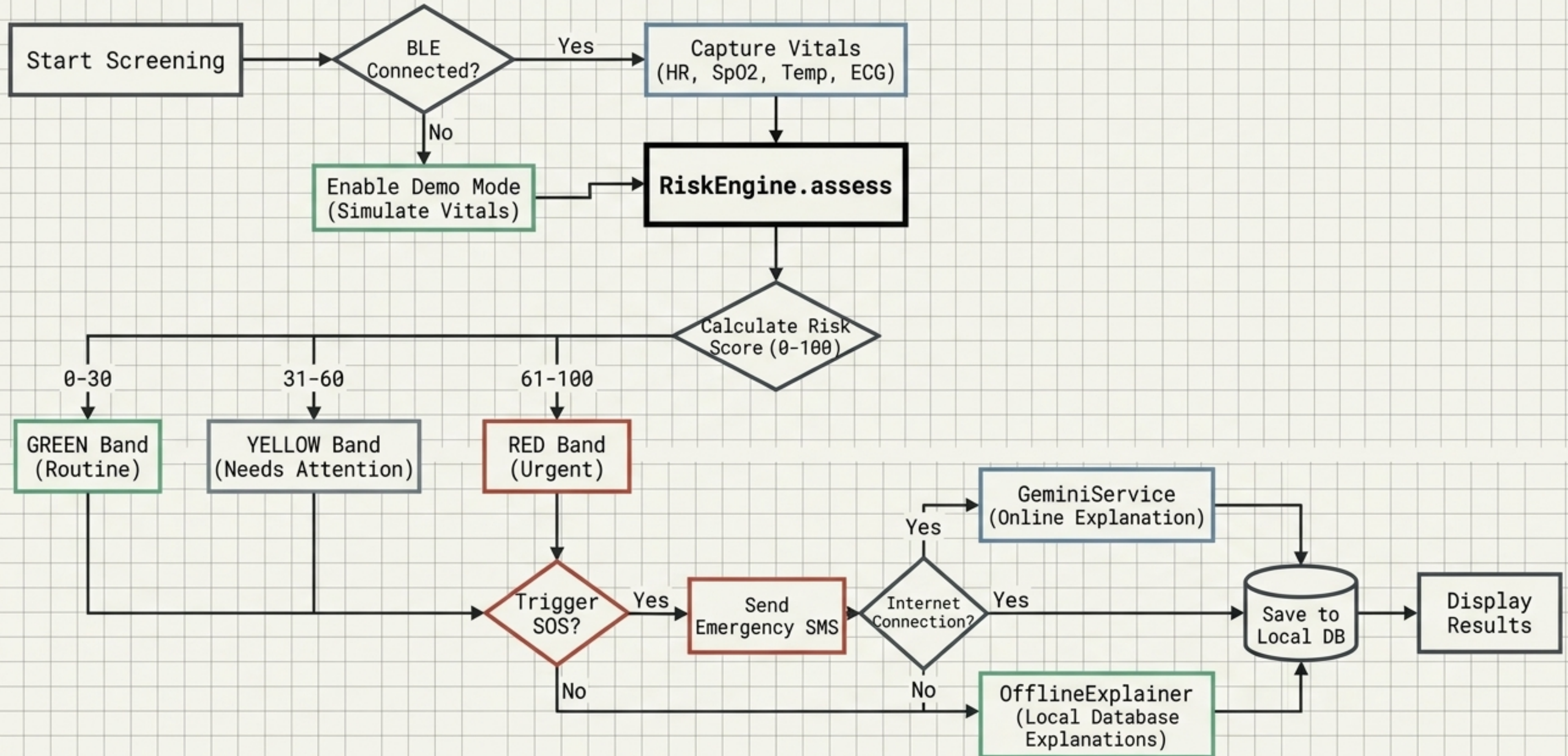
Hardware Testing & Validation Protocol

```
> INIT PHASE 1: CONTINUITY CHECKS
[ OK ] GND pins connected.
[ OK ] 3.3V rail stable (Measuring 3.25-3.35V).
[ OK ] I2C SDA/SCL 4.7kΩ pull-ups active.

> INIT PHASE 2: FIRMWARE SELF-TEST (1-11)
[ OK ] ESP32 Boot
[ OK ] I2C Scan (3+ devices detected)
[ OK ] MAX30102 detected (0x57)
[ OK ] MLX90614 detected (0x5A)
[ OK ] AD8232 signal in range
[ OK ] L0+/L0- connection validated
[ OK ] BLE advertising active
```

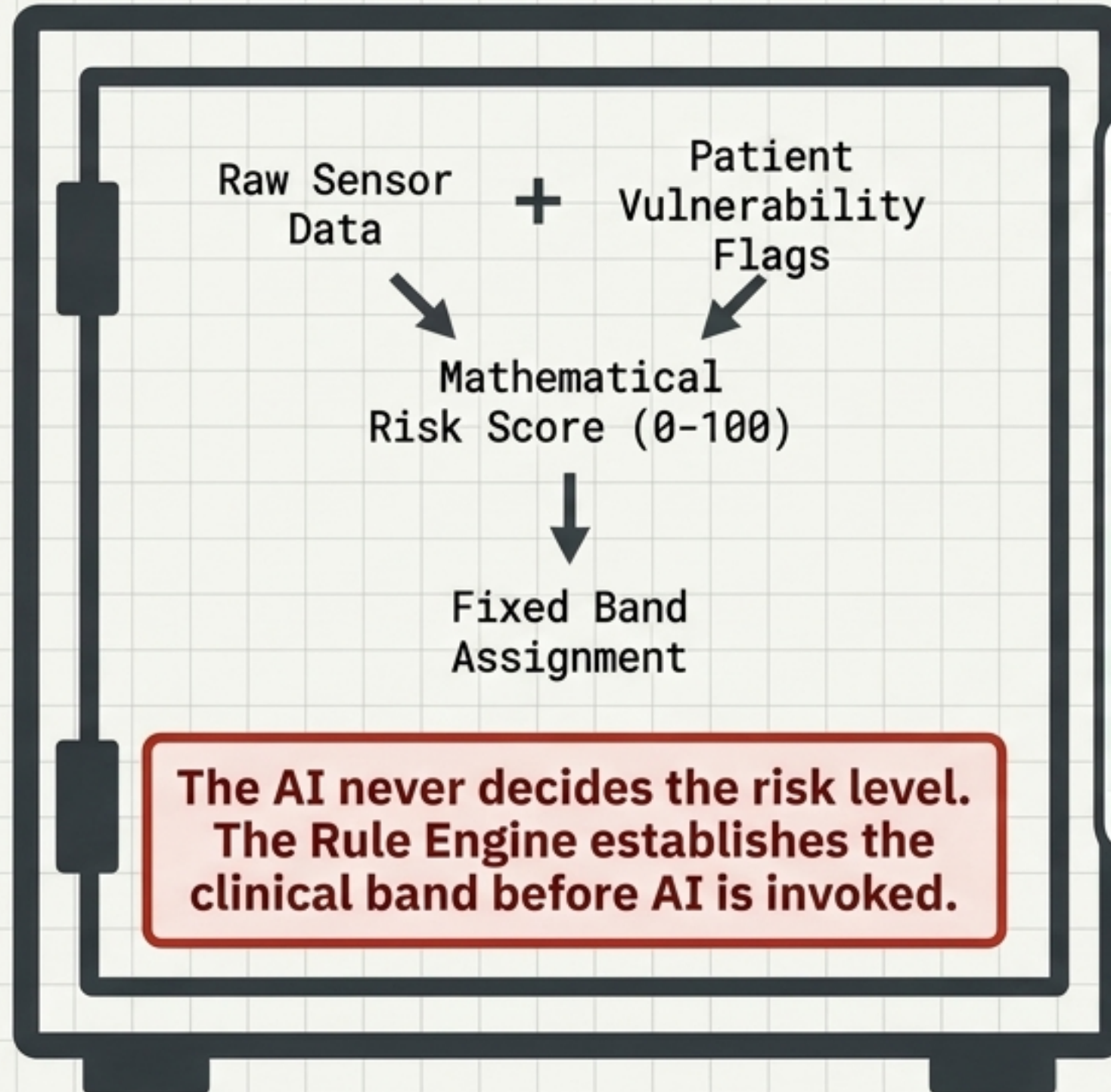


End-to-End Application Screening Lifecycle



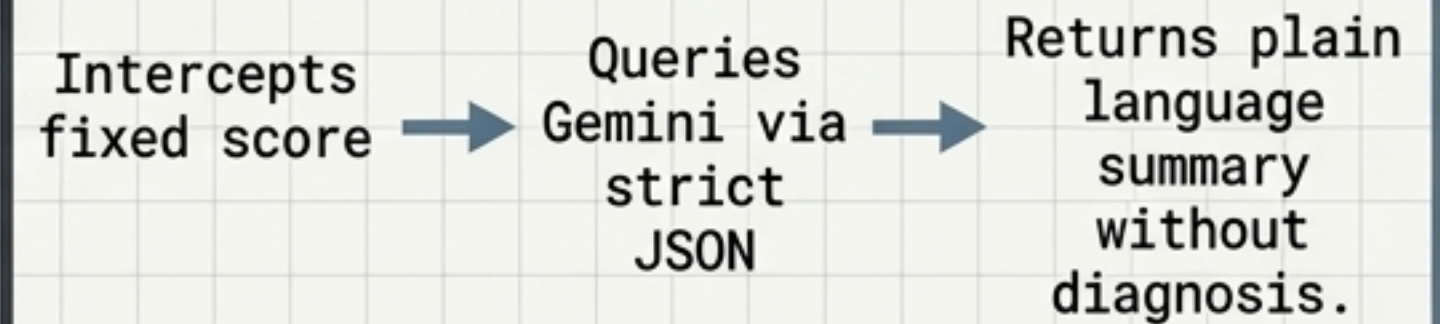
Dual-Tier Explanation & Security Architecture

The Safety Boundary (Deterministic)

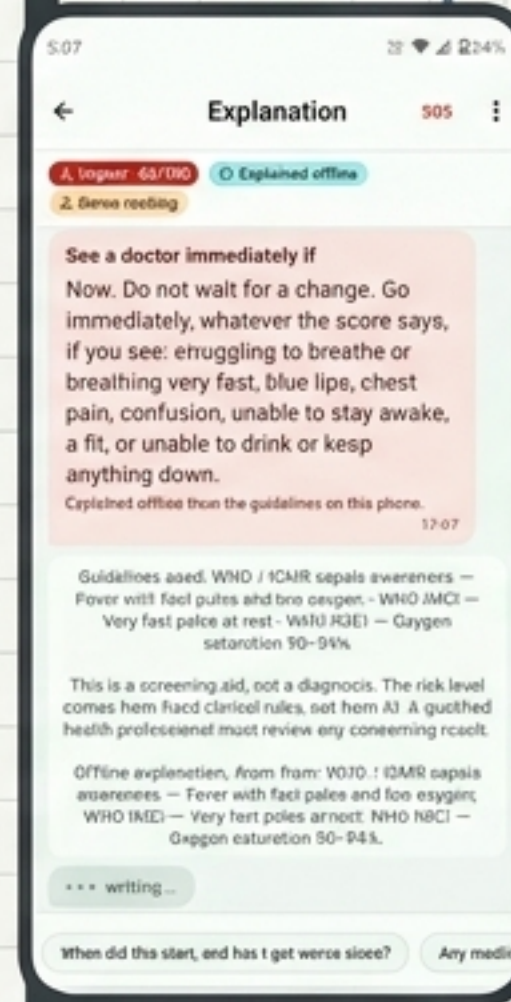
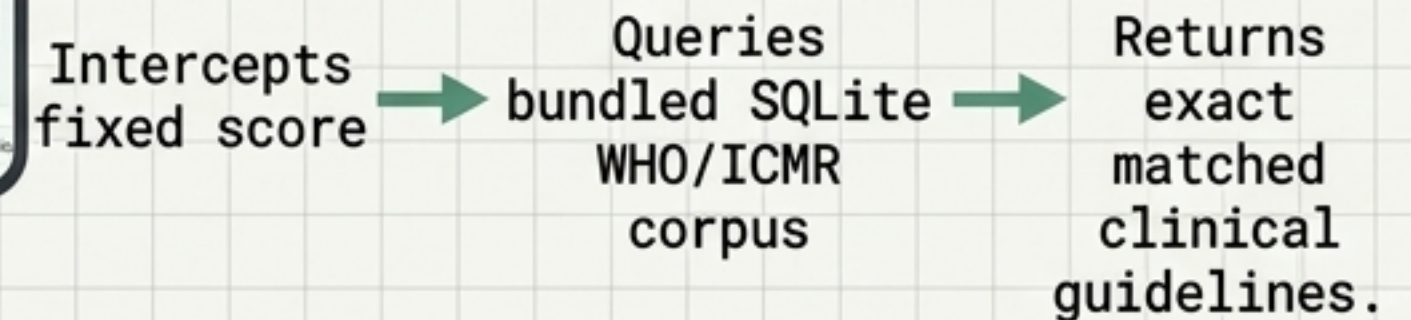


Multi-Tier Explanation

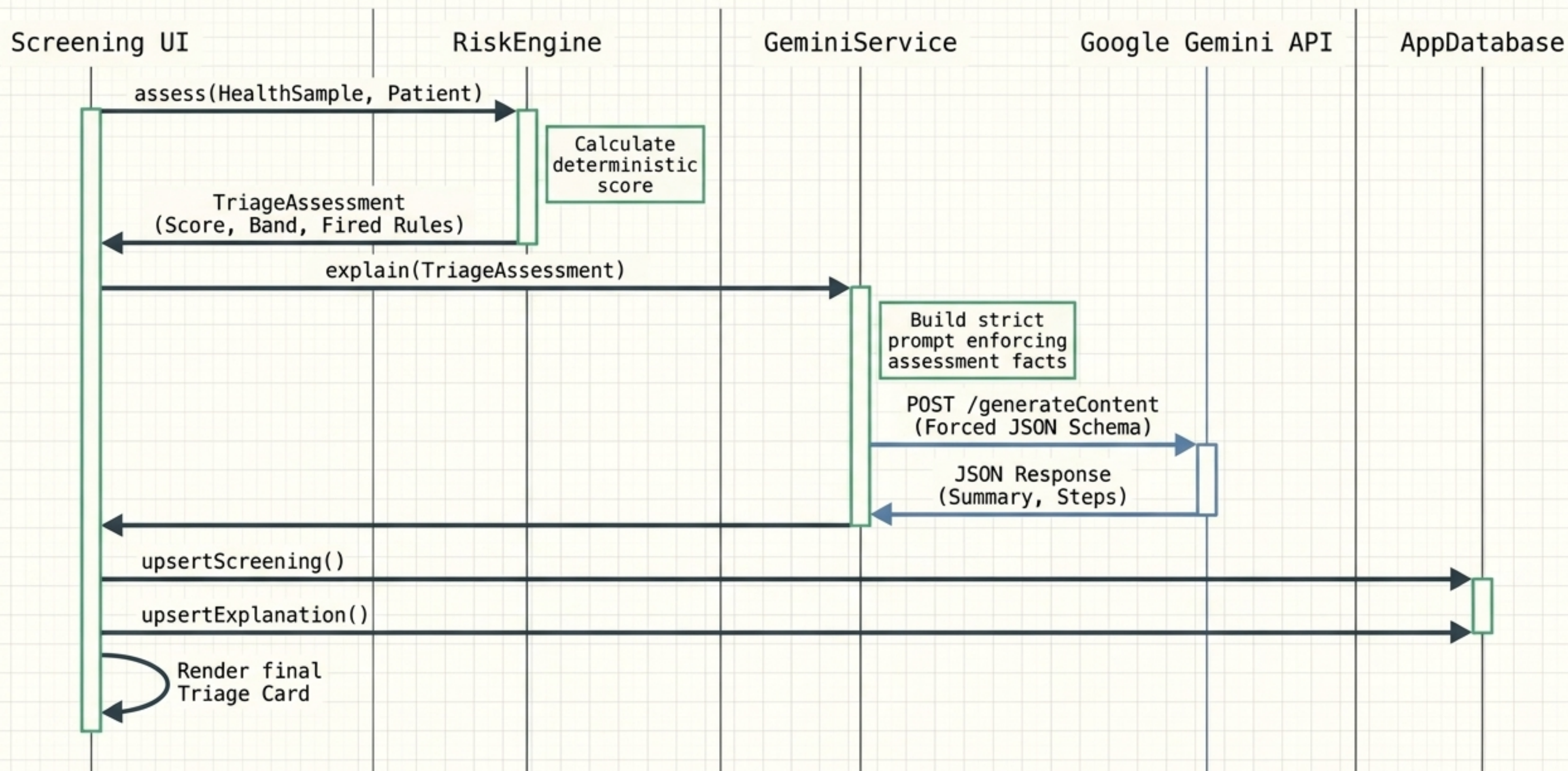
Tier 1 (Online)



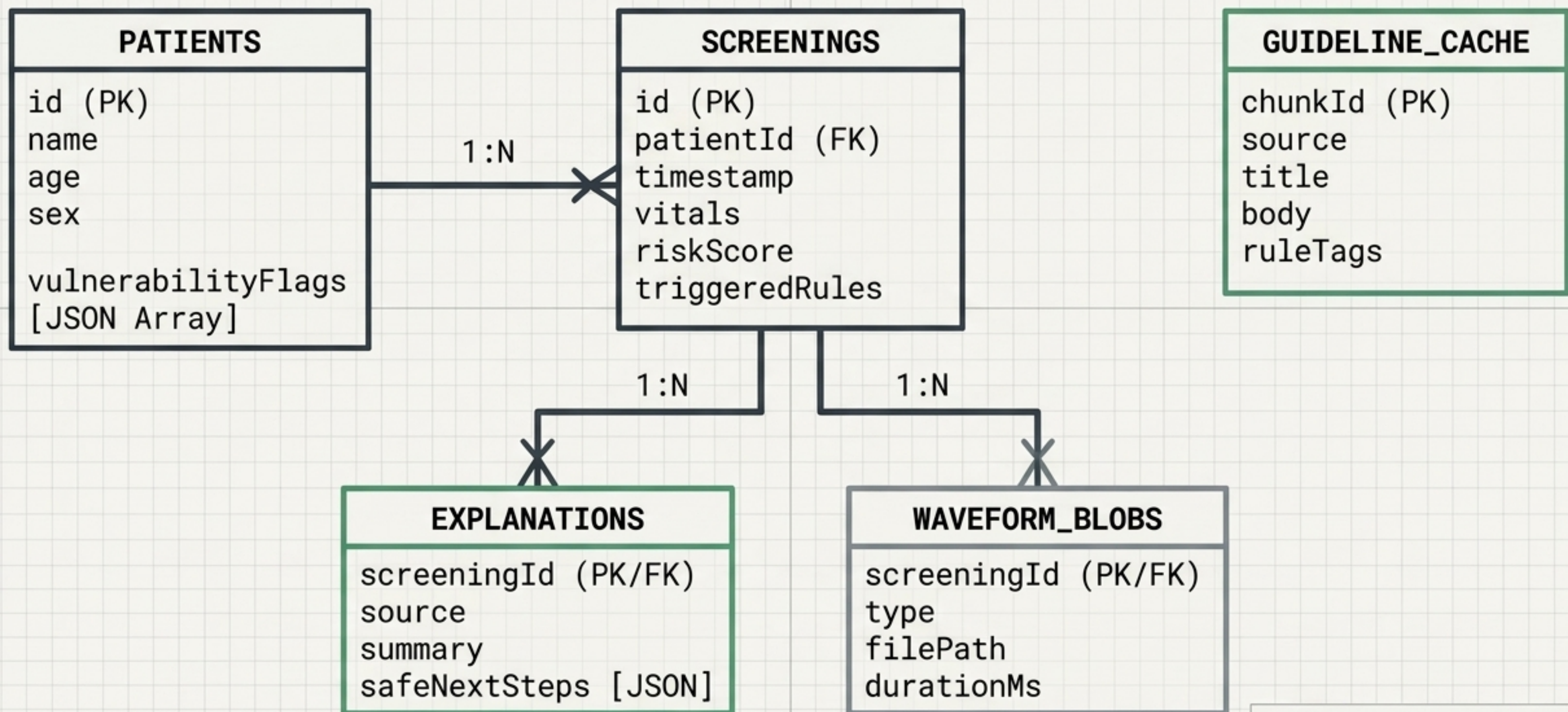
Tier 2 (Offline)



AI Explanation Generation Sequence (JSON Constrained)

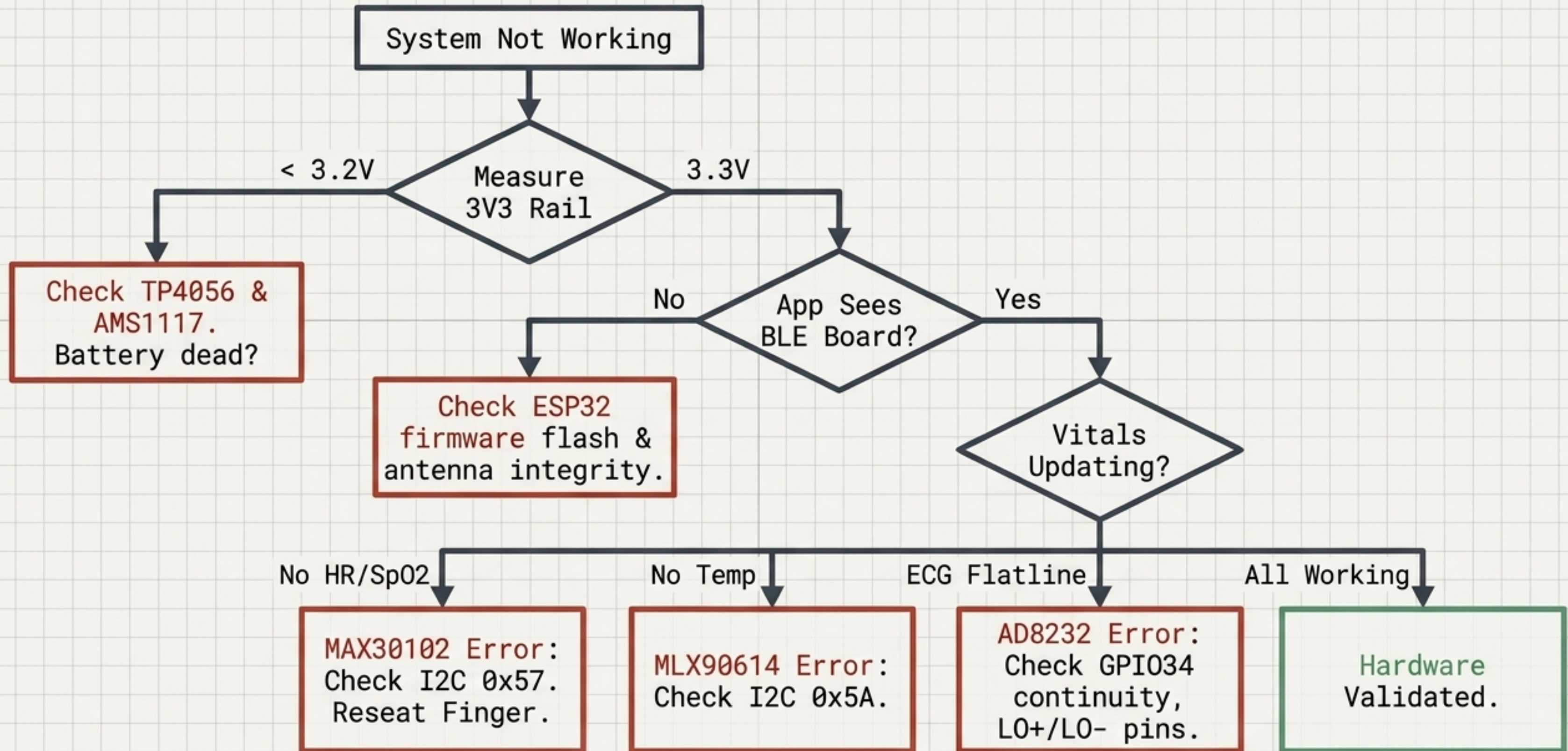


Offline Database Schema: Zero-Dependency Architecture



Implementation: Drift SQLite
(Offline-First Storage Layer)

End-to-End Hardware Troubleshooting Matrix



Appendix: Open Source Engineering Repository

Open Source Verification

Repository:

github.com/krishnendukoley2007-arch/swasthyasetu-ai

[QR CODE]

Core Technical Stack & QA

Edge Firmware: ESP-IDF / Arduino C++

Application: Flutter 3.47+ / Dart 3.13+

Database: Drift SQLite

Validation: 428 passing unit/widget tests validating deterministic rules and UI overflow bounds.

Medical Disclaimer

SwasthyaSetu AI is a screening aid, not a diagnosis. The rule engine applies fixed WHO/ICMR criteria. AI is strictly constrained to explanations and cannot override the deterministic math. Do not use as the sole basis for critical care.